Lab 03

TITLE

Working with Git History, Stashing, and Reverting Commits

Name:

Areeba Hameed

Registration number:

(2023-BSCS-15)

Section:

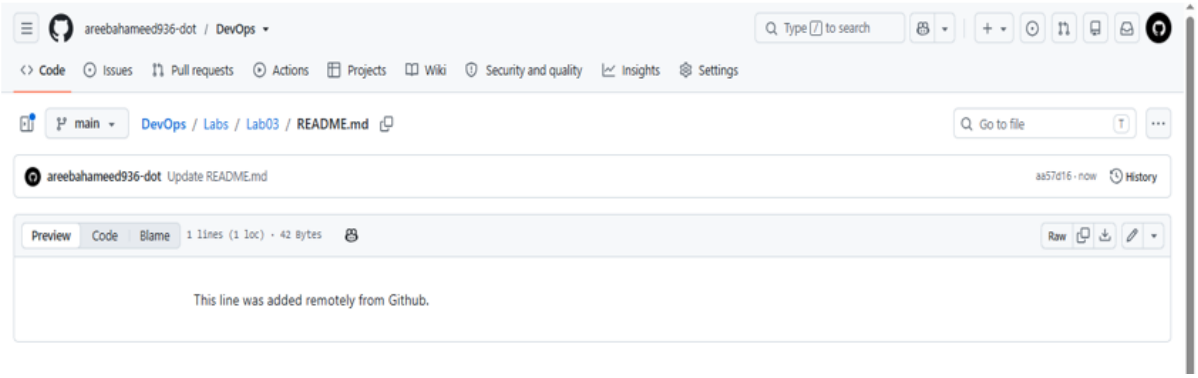
A

GitHub username:

Areebahameed936-dot

Task 1 – Handling Local and Remote Commit Conflicts (Pull vs Pull --rebase)

remote_edit.png



local_commit.png

```
areebahameed936-dot@ubunto:~$ cd ~/DevOps
areebahameed936-dot@ubunto:~/DevOps$ echo "This line was added locally" >> README.md
areebahameed936-dot@ubunto:~/DevOps$ git add README.md
areebahameed936-dot@ubunto:~/DevOps$ git commit -m "Local update to README"
[main 9c1af0c] Local update to README
 1 file changed, 1 insertion(+)
areebahameed936-dot@ubunto:~/DevOps$
```

push_error.png

```
areebahameed936-dot@ubunto:~/DevOps$ git push origin main
To github.com:areebahameed936-dot/DevOps.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:areebahameed936-dot/DevOps.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
areebahameed936-dot@ubunto:~/DevOps$
```

merge_commit.png

```
Merge made by the 'ort' strategy.
 Labs/Lab02/Lab02_Solution.pdf           | Bin 0 -> 1268514 bytes
 Labs/Lab03/Lab3_Solution.pdf            |   1 +
 Labs/Lab03/README.md                    |   1 +
 Labs/Lab03/compose.yaml                 |   1 +
 Labs/Lab03/screenshots/local_commit.png |   1 +
 Labs/Lab03/screenshots/remote_edit.png  |   1 +
 Labs/Lab03/temp.txt                     |   1 +
 7 files changed, 6 insertions(+)
 create mode 100644 Labs/Lab02/Lab02_Solution.pdf
 create mode 100644 Labs/Lab03/Lab3_Solution.pdf
 create mode 100644 Labs/Lab03/README.md
 create mode 100644 Labs/Lab03/compose.yaml
 create mode 100644 Labs/Lab03/screenshots/local_commit.png
 create mode 100644 Labs/Lab03/screenshots/remote_edit.png
 create mode 100644 Labs/Lab03/temp.txt
areebahameed936-dot@ubunto:~/DevOps$
```

push_after_merge.png

```
areebahameed936-dot@ubunto:~/DevOps$ git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (8/8), done.
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 666 bytes | 222.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:areebahameed936-dot/DevOps.git
   aa57d16..4cfde0e  main -> main
branch 'main' set up to track 'origin/main'.
```

rebase_pull.png

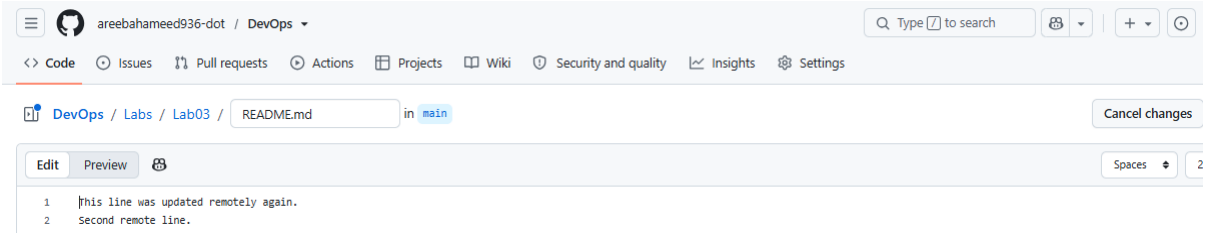
```
areebahameed936-dot@ubunto:~/DevOps$ git pull --rebase origin main
From github.com:areebahameed936-dot/DevOps
 * branch            main       -> FETCH_HEAD
Already up to date.
areebahameed936-dot@ubunto:~/DevOps$
```

push_after_rebase.png

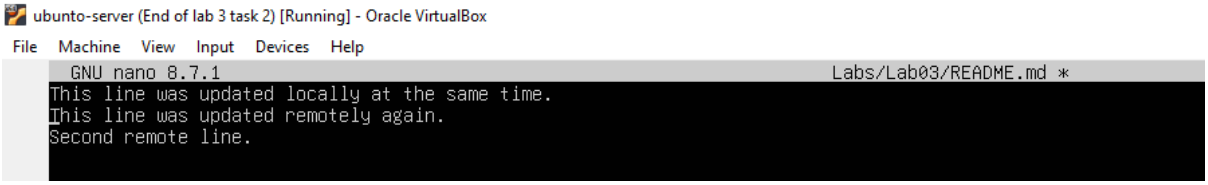
```
areebahameed936-dot@ubunto:~/DevOps$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
areebahameed936-dot@ubunto:~/DevOps$ _
```

Task 2 – Creating and Resolving Merge Conflicts Manually

remote_conflict_edit.png



local_conflict_edit.png



local_conflict_commit.png

```
areebahameed936-dot@ubunto:~/DevOps$ git add Labs/Lab03/README.md
areebahameed936-dot@ubunto:~/DevOps$ git commit -m "Local conflicting change"
[main 7f1f3e3] Local conflicting change
 1 file changed, 2 insertions(+), 1 deletion(-)
areebahameed936-dot@ubunto:~/DevOps$ _
```

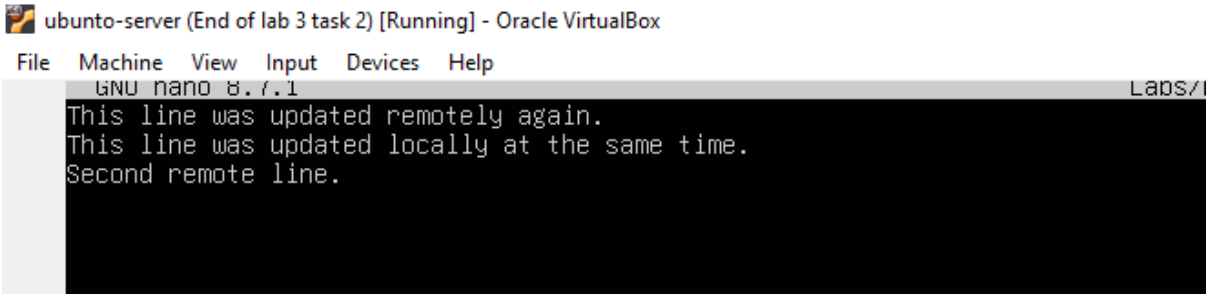
conflict_push_error.png

```
areebahameed936-dot@ubunto:~/DevOps$ git push -u origin main
To github.com:areebahameed936-dot/DevOps.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:areebahameed936-dot/DevOps.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
areebahameed936-dot@ubunto:~/DevOps$
```

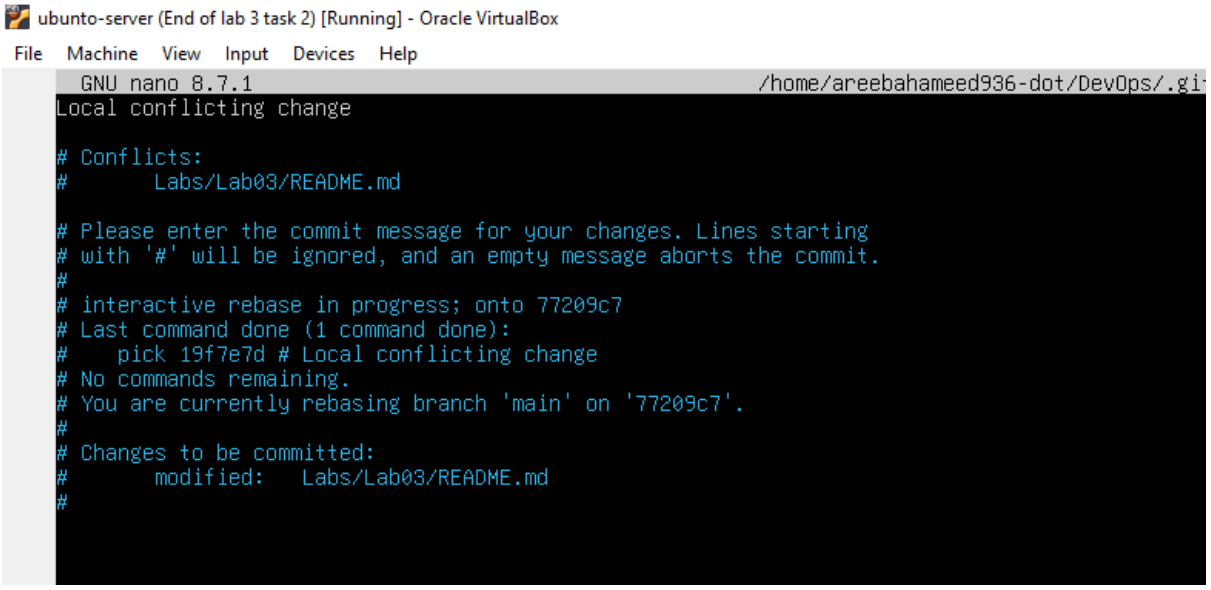
conflict_message.png

```
areebahameed936-dot@ubunto:~/DevOps$ git pull --rebase origin main
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.04 KiB | 178.00 KiB/s, done.
From github.com:areebahameed936-dot/DevOps
 * branch            main       -> FETCH_HEAD
   e9c4750..77209c7  main       -> origin/main
Auto-merging Labs/Lab03/README.md
CONFLICT (content): Merge conflict in Labs/Lab03/README.md
error: could not apply 19f7e7d... Local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply 19f7e7d... # Local conflicting change
areebahameed936-dot@ubunto:~/DevOps$ _
```

resolved_readme.png



rebase_continue.png

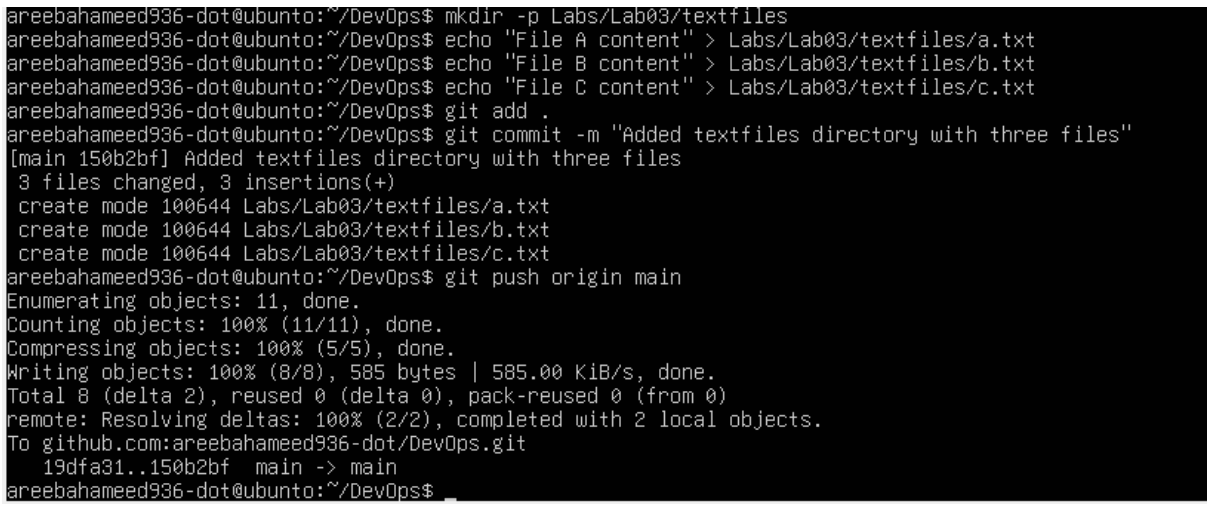


push_after_resolve.png



Task 3 –Managing Ignored Files with .gitignore and Removing Tracked Files

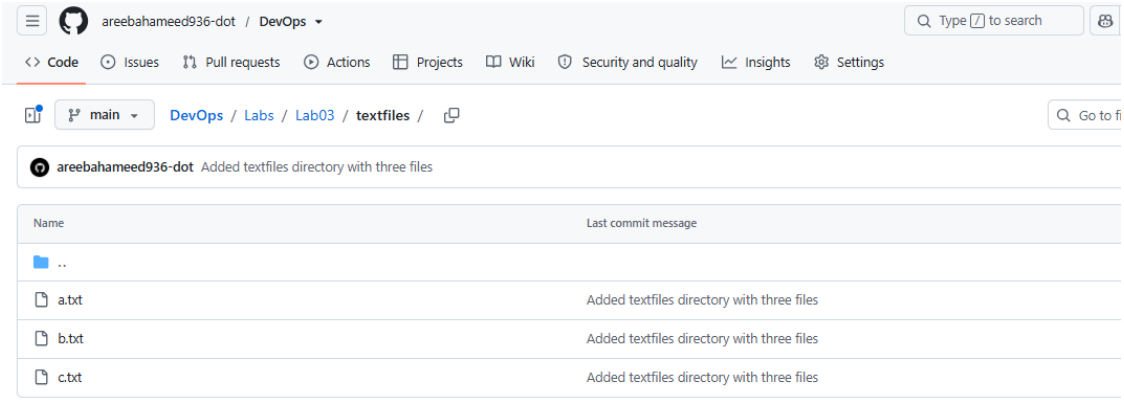
push_textfiles.png



gitignore_push.png

```
areebahameed936-dot@ubunto:~/DevOps$ echo "textfiles/*" > Labs/Lab03/.git ignore
areebahameed936-dot@ubunto:~/DevOps$ echo "textfiles/*" > Labs/Lab03/.gitignore
areebahameed936-dot@ubunto:~/DevOps$ git push origin main
Everything up-to-date
areebahameed936-dot@ubunto:~/DevOps$
```

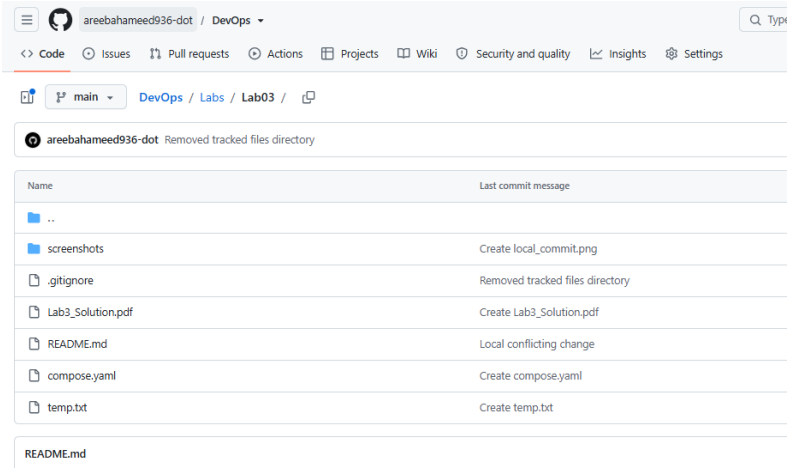
repo_still_has_textfiles.png



rm_cached_push.png

```
areebahameed936-dot@ubunto:~/DevOps$ git rm -r --cached Labs/Lab03/textfiles
rm 'Labs/Lab03/textfiles/a.txt'
rm 'Labs/Lab03/textfiles/b.txt'
rm 'Labs/Lab03/textfiles/c.txt'
areebahameed936-dot@ubunto:~/DevOps$ git add .
areebahameed936-dot@ubunto:~/DevOps$ git commit -m "Removed tracked files directory"
[main a060f28] Removed tracked files directory
 4 files changed, 1 insertion(+), 3 deletions(-)
 create mode 100644 Labs/Lab03/.gitignore
 delete mode 100644 Labs/Lab03/textfiles/a.txt
 delete mode 100644 Labs/Lab03/textfiles/b.txt
 delete mode 100644 Labs/Lab03/textfiles/c.txt
areebahameed936-dot@ubunto:~/DevOps$ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 439 bytes | 146.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:areebahameed936-dot/DevOps.git
   150b2bf..a060f28  main -> main
areebahameed936-dot@ubunto:~/DevOps$
```

repo_textfiles_removed.png



Task 4 –Create Temporary Changes and Use git stash

modified_readme.png

```
areebahameed936-dot@ubunto:~/DevOps$ git checkout -b feature-branch
fatal: a branch named 'feature-branch' already exists
areebahameed936-dot@ubunto:~/DevOps$ echo "Test line 1 for stash task" >> Labs/Lab03/README.md
areebahameed936-dot@ubunto:~/DevOps$ git add Labs/Lab03/README.md
areebahameed936-dot@ubunto:~/DevOps$ git commit -m "Changes the README.md file."
[main da92d9a] Changes the README.md file.
 1 file changed, 2 insertions(+)
areebahameed936-dot@ubunto:~/DevOps$
```

checkout_error.png

```
areebahameed936-dot@ubunto:~/DevOps$ echo "Test line 2 - Uncommited change." >> Labs/Lab03/README.md
areebahameed936-dot@ubunto:~/DevOps$ git checkout main
M       Labs/Lab03/README.md
Already on 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
```

stash_command.png

```
areebahameed936-dot@ubunto:~/DevOps$ git stash
Saved working directory and index state WIP on main: da92d9a Changes the README.md file.
areebahameed936-dot@ubunto:~/DevOps$
```

branch_switched.png

```
areebahameed936-dot@ubunto:~/DevOps$ git checkout main
M       Labs/Lab03/README.md
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

back_to_feature.png

```
areebahameed936-dot@ubunto:~/DevOps$ git checkout feature-branch
A       Labs/Lab03?README.md
Switched to branch 'feature-branch'
areebahameed936-dot@ubunto:~/DevOps$
```

status_clean.png

```
areebahameed936-dot@ubunto:~/DevOps$ git status
On branch feature-branch
nothing to commit, working tree clean
areebahameed936-dot@ubunto:~/DevOps$
```

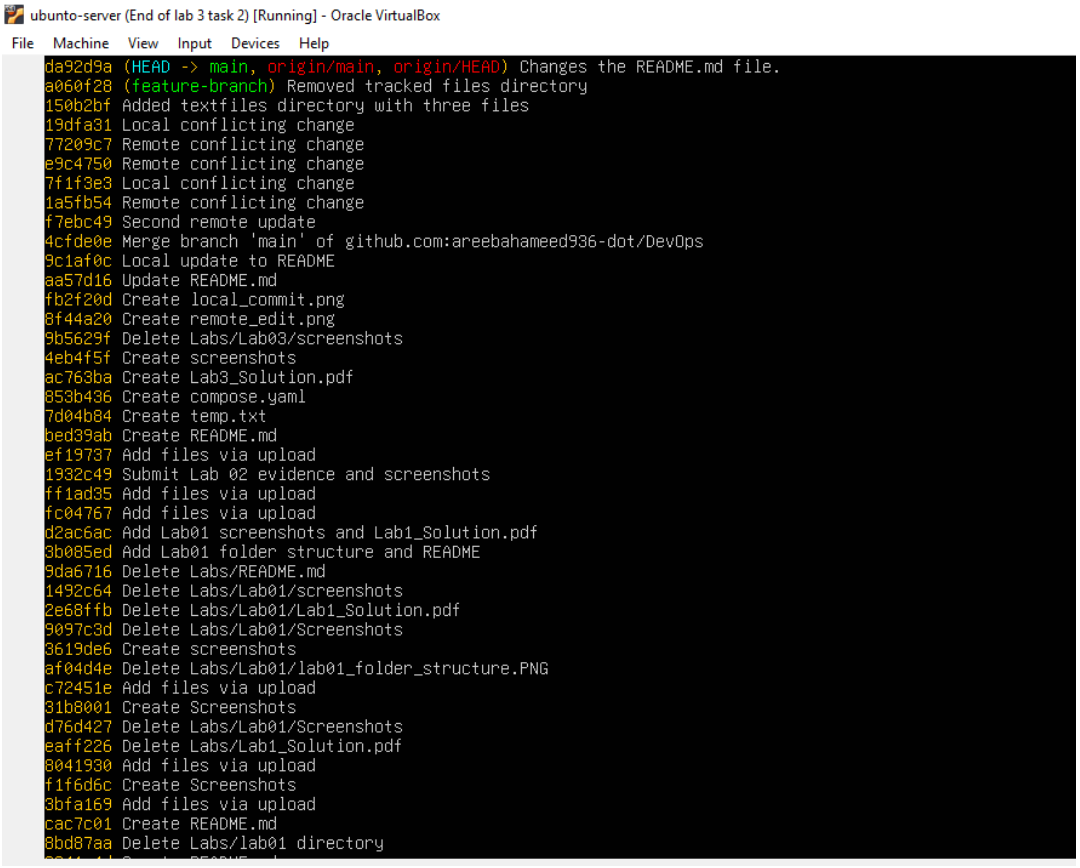
stash_pop.png

```
areebahameed936-dot@ubunto:~/DevOps$ git stash pop
Auto-merging Labs/Lab03/README.md
CONFLICT (content): Merge conflict in Labs/Lab03/README.md
On branch feature-branch
Unmerged paths:
  (use "git restore --staged <file>..." to unstage)
  (use "git add <file>..." to mark resolution)
        both modified:   Labs/Lab03/README.md

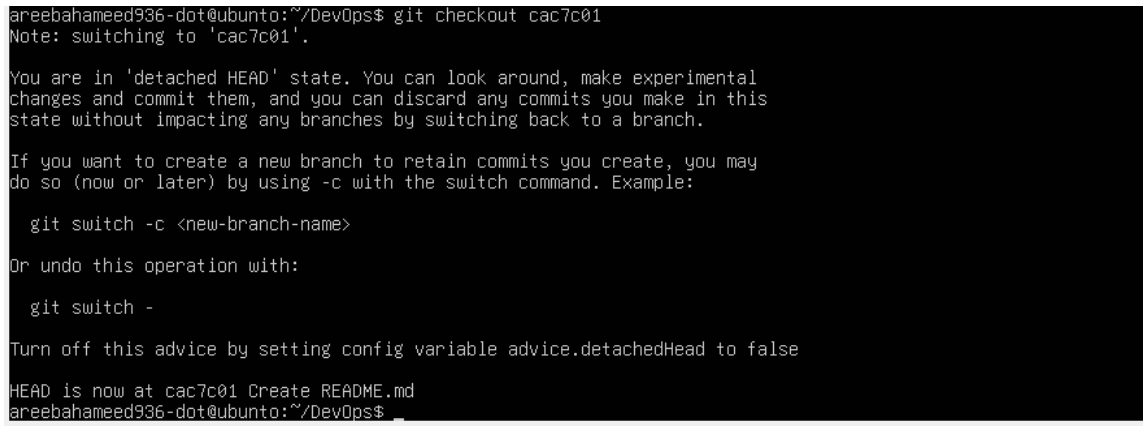
no changes added to commit (use "git add" and/or "git commit -a")
The stash entry is kept in case you need it again.
areebahameed936-dot@ubunto:~/DevOps$ _
```

Task 5 – Checkout a Specific Commit Using git log

log_before_checkout.png



detached_head.png

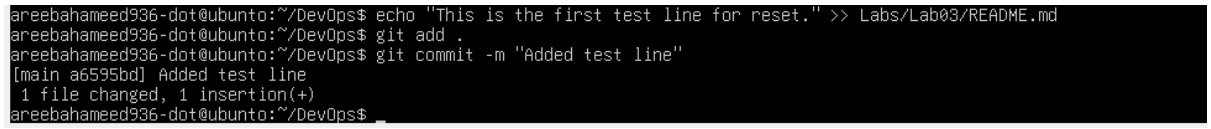


back_to_main.png

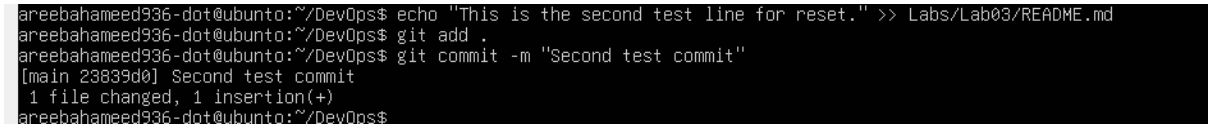


Task 6 – Resetting Commits (Soft vs Hard Reset) (With Verification Steps)

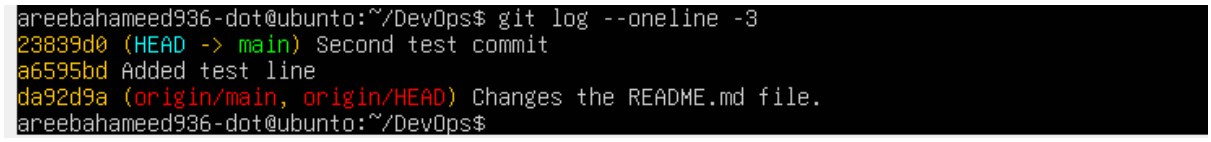
first_commit.png



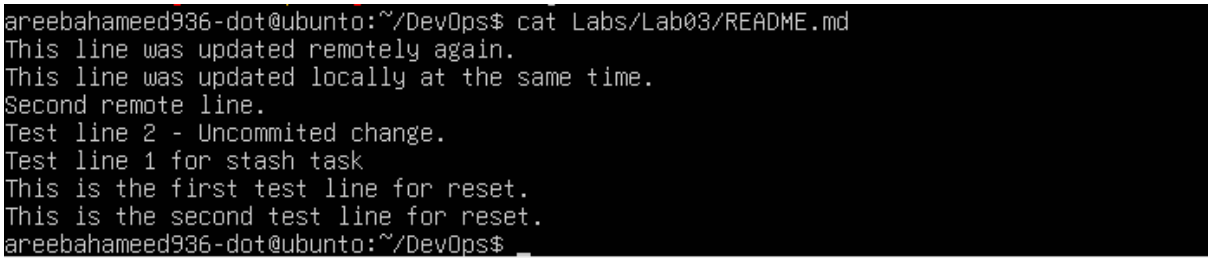
second_commit.png



log_before_reset.png



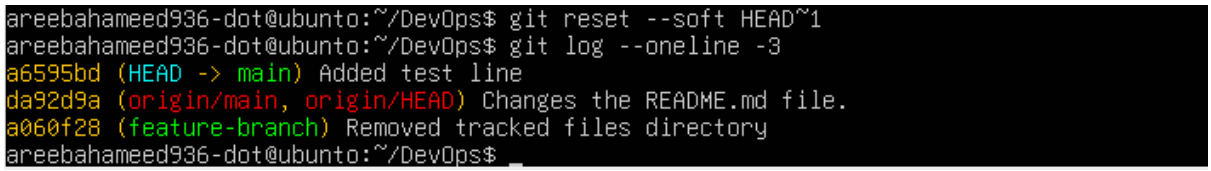
file_before_reset.png



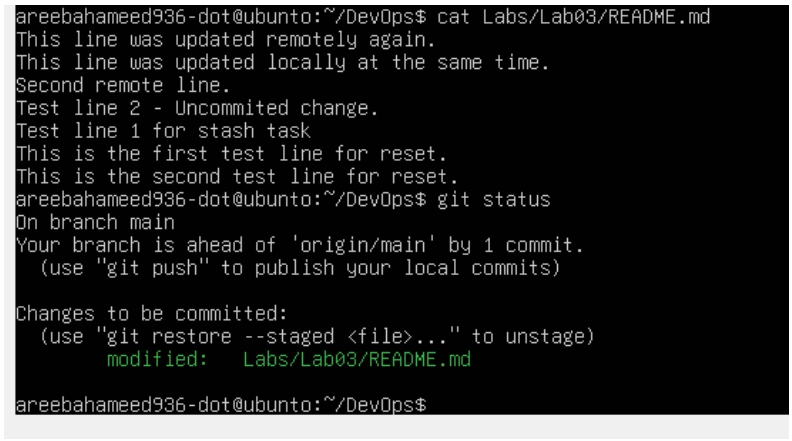
soft_reset.png



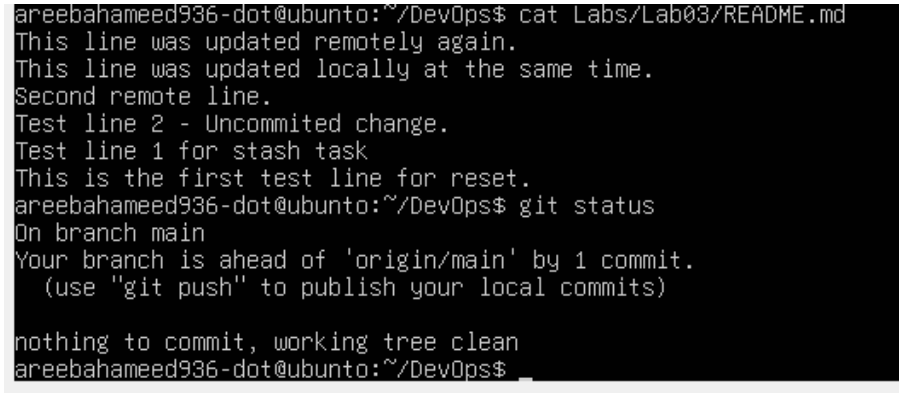
log_after_soft_reset.png



file_after_soft_reset.png



file_after_hard_reset.png



hard_reset.png

```
areebahameed936-dot@ubunto:~/DevOps$ git reset --hard HEAD~1
HEAD is now at a6595bd Added test line
areebahameed936-dot@ubunto:~/DevOps$
```

log_after_hard_reset.png

```
areebahameed936-dot@ubunto:~/DevOps$ git log --oneline -3
a6595bd (HEAD -> main) Added test line
da92d9a (origin/main, origin/HEAD) Changes the README.md file.
a060f28 (feature-branch) Removed tracked files directory
areebahameed936-dot@ubunto:~/DevOps$
```

file_after_hard_reset.png

```
areebahameed936-dot@ubunto:~/DevOps$ cat Labs/Lab03/README.md
This line was updated remotely again.
This line was updated locally at the same time.
Second remote line.
Test line 2 - Uncommited change.
Test line 1 for stash task
This is the first test line for reset.
areebahameed936-dot@ubunto:~/DevOps$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
areebahameed936-dot@ubunto:~/DevOps$
```

Task 7 – Amending the Last Commit

first_amend_commit.png

```
areebahameed936-dot@ubunto:~/DevOps$ echo "This is an incomplete commit" >> Labs/Lab03/temp.txt
areebahameed936-dot@ubunto:~/DevOps$ git add .
areebahameed936-dot@ubunto:~/DevOps$ git commit -m "Fix log message"
[main 2e3de75] Fix log message
 1 file changed, 1 insertion(+)
areebahameed936-dot@ubunto:~/DevOps$
```

amend_commit.png

```
[main 0bb6e88] Fix log message
 Date: Tue Oct 20 16:14:36 2026 +0000
 2 files changed, 2 insertions(+)
areebahameed936-dot@ubunto:~/DevOps$
```

Task 8 – Reverting a Commit (Safe Undo on Remote Branch)

commit_temp_file.png

```
areebahameed936-dot@ubunto:~/DevOps$ echo "temporary text" >> Labs/Lab03/temp.txt
areebahameed936-dot@ubunto:~/DevOps$ git add .
areebahameed936-dot@ubunto:~/DevOps$ git commit -m "Added temporary text file"
[main 10cfc9f] Added temporary text file
 1 file changed, 1 insertion(+)
areebahameed936-dot@ubunto:~/DevOps$ git push origin main
Enumerating objects: 22, done.
Counting objects: 100% (21/21), done.
Compressing objects: 100% (13/13), done.
Writing objects: 100% (16/16), 1.31 KiB | 78.00 KiB/s, done.
Total 16 (delta 7), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (7/7), completed with 3 local objects.
To github.com:areebahameed936-dot/DevOps.git
   da92d9a..10cfc9f  main -> main
areebahameed936-dot@ubunto:~/DevOps$
```

revert_commit.png

```
areebahameed936-dot@ubunto:~/DevOps$ git log --oneline -3
10cfc9f (HEAD -> main, origin/main, origin/HEAD) Added temporary text file
0bb6e88 Fix log message
a6595bd Added test line
areebahameed936-dot@ubunto:~/DevOps$
```

revert_push.png

```
areebahameed936-dot@ubunto:~/DevOps$ git push origin main
Everything up-to-date
areebahameed936-dot@ubunto:~/DevOps$
```

Task 9 – Force Push (With Caution)

new_branch.png

```
areebahameed936-dot@ubunto:~/DevOps$ git checkout -b test-force
Switched to a new branch 'test-force'
areebahameed936-dot@ubunto:~/DevOps$
```

force_commit.png

```
areebahameed936-dot@ubunto:~/DevOps$ echo "Force push test" >> Labs/Lab03/temp.txt
areebahameed936-dot@ubunto:~/DevOps$ git add .
areebahameed936-dot@ubunto:~/DevOps$ git commit -m "Force push test commit"
[test-force 53e781a] Force push test commit
 1 file changed, 1 insertion(+)
areebahameed936-dot@ubunto:~/DevOps$
```

push_force_branch.png

```
areebahameed936-dot@ubunto:~/DevOps$ git push origin test-force
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 455 bytes | 26.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/areebahameed936-dot/DevOps/pull/new/test-force
remote:
To github.com:areebahameed936-dot/DevOps.git
 * [new branch]      test-force -> test-force
areebahameed936-dot@ubunto:~/DevOps$
```

hard_reset_force.png

```
areebahameed936-dot@ubunto:~/DevOps$ git reset --hard HEAD~1
HEAD is now at 10cfc9f Added temporary text file
areebahameed936-dot@ubunto:~/DevOps$
```

normal_push.png

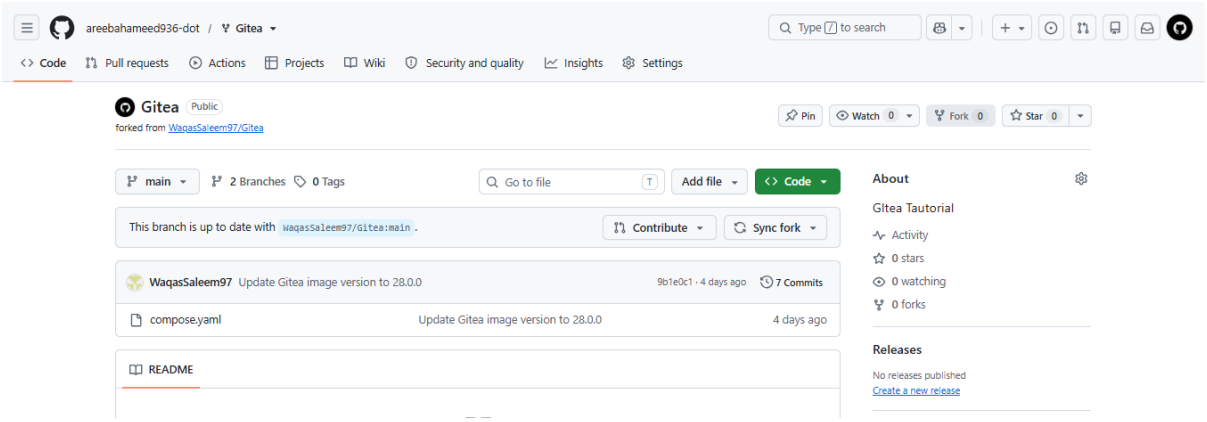
```
areebahameed936-dot@ubunto:~/DevOps$ git push origin test-force
To github.com:areebahameed936-dot/DevOps.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'github.com:areebahameed936-dot/DevOps.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
areebahameed936-dot@ubunto:~/DevOps$
```

force_push.png

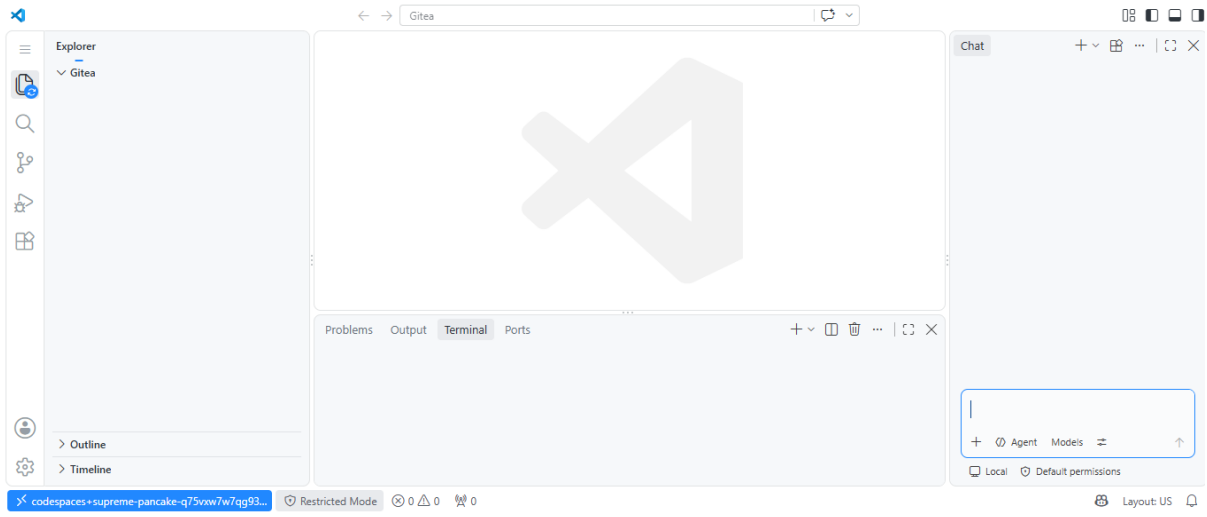
```
areebahameed936-dot@ubunto:~/DevOps$ git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:areebahameed936-dot/DevOps.git
 + 53e781a...10cfc9f test-force -> test-force (forced update)
areebahameed936-dot@ubunto:~/DevOps$
```

Task 10 – Running Gitea in GitHub Codespaces via Docker Compose

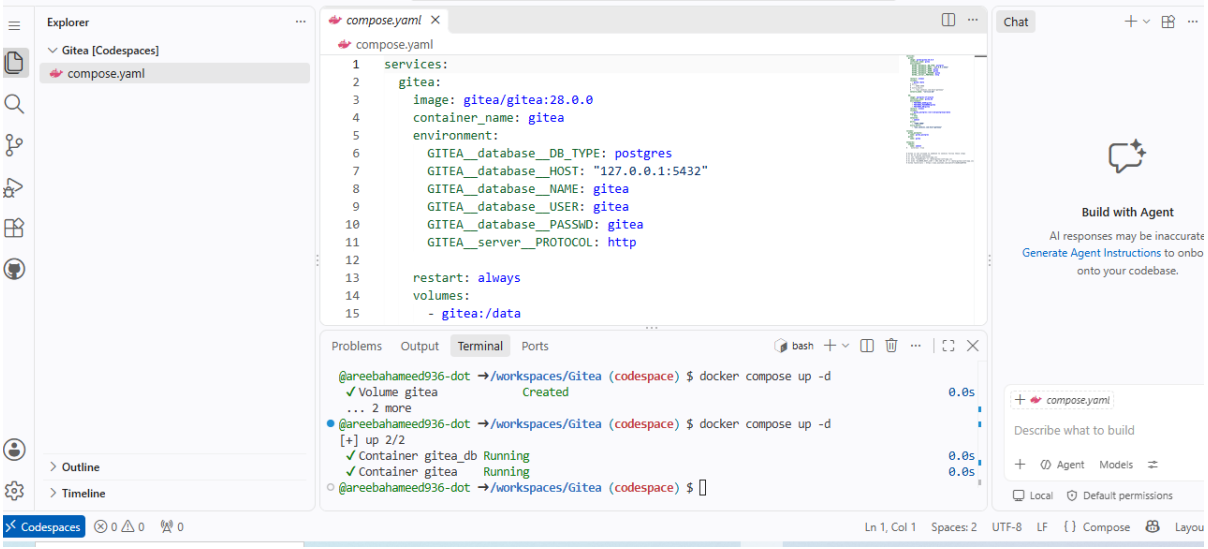
forked_gitea.png



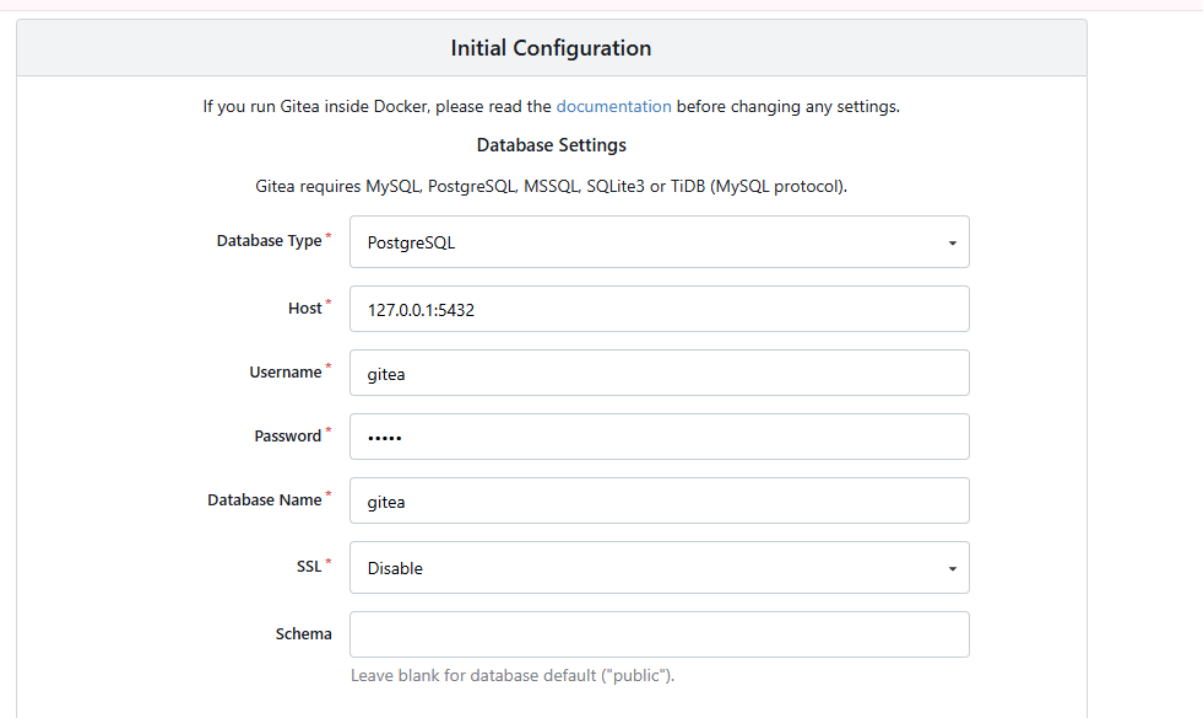
codespace_loading.png



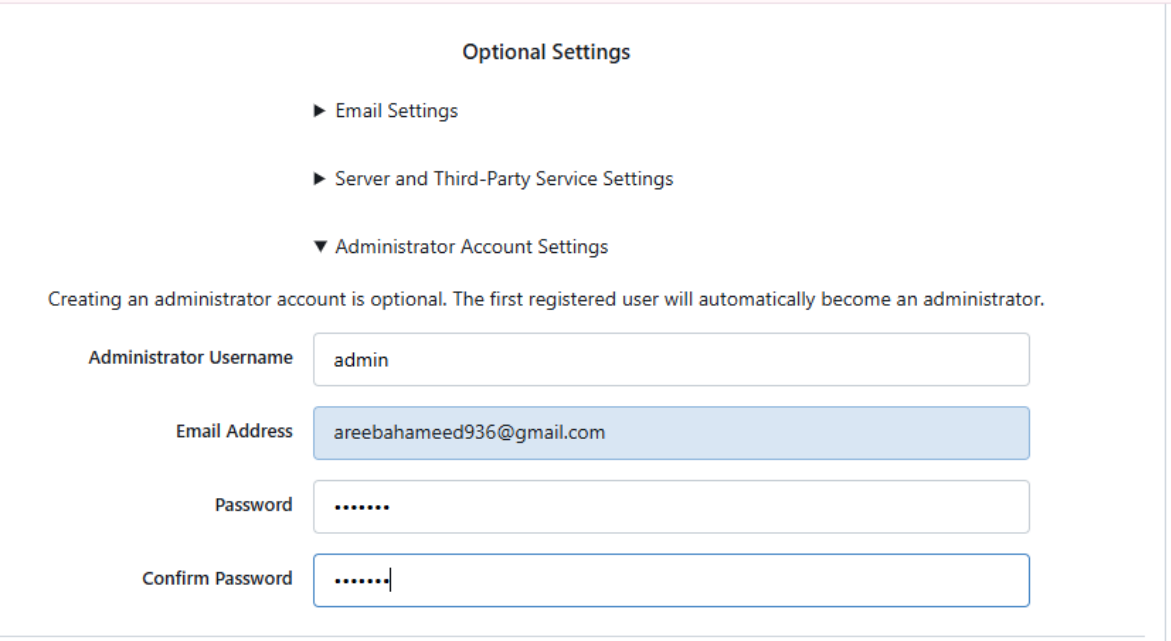
docker_up.png



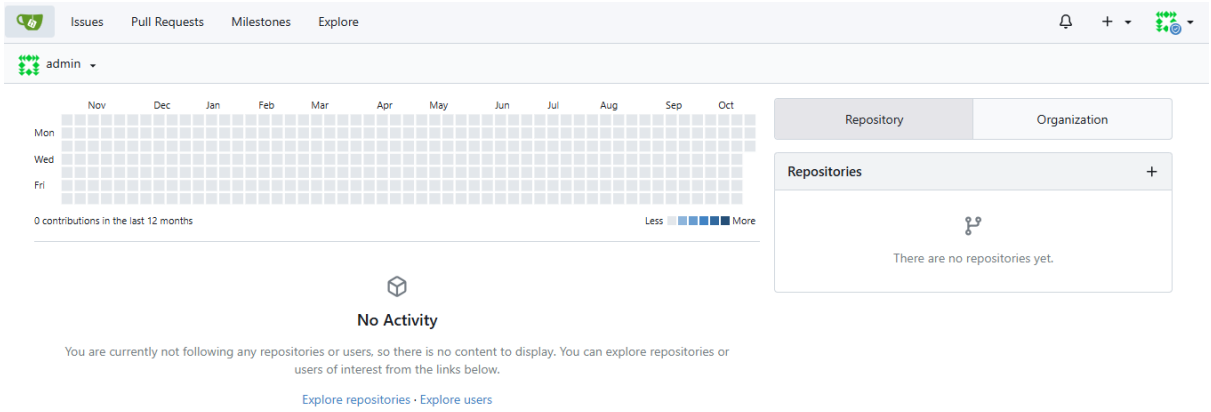
gitea_install_page.png



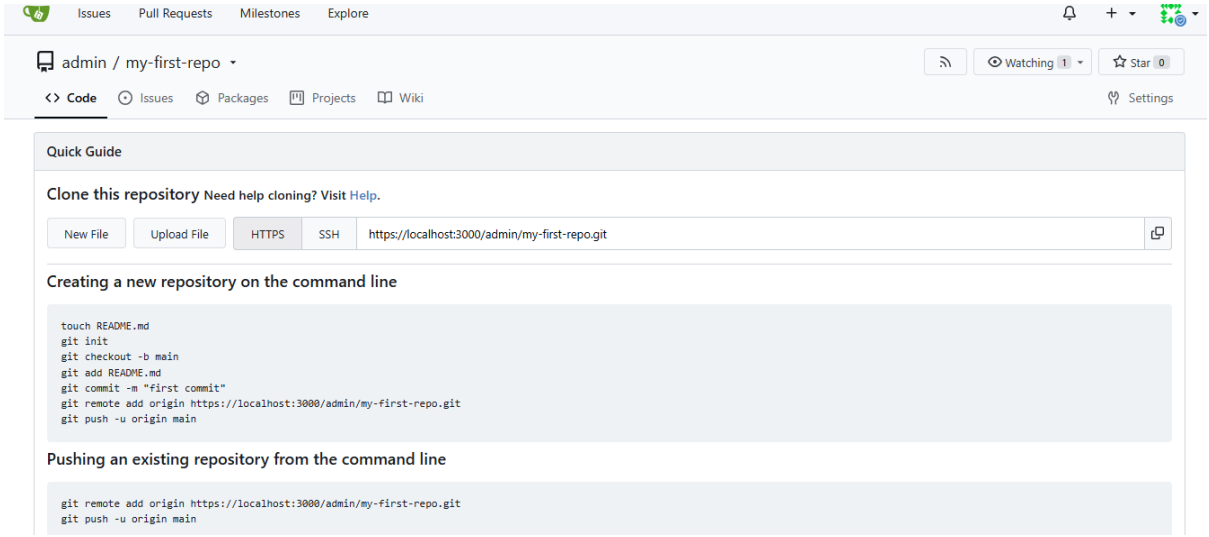
admin_setup.png



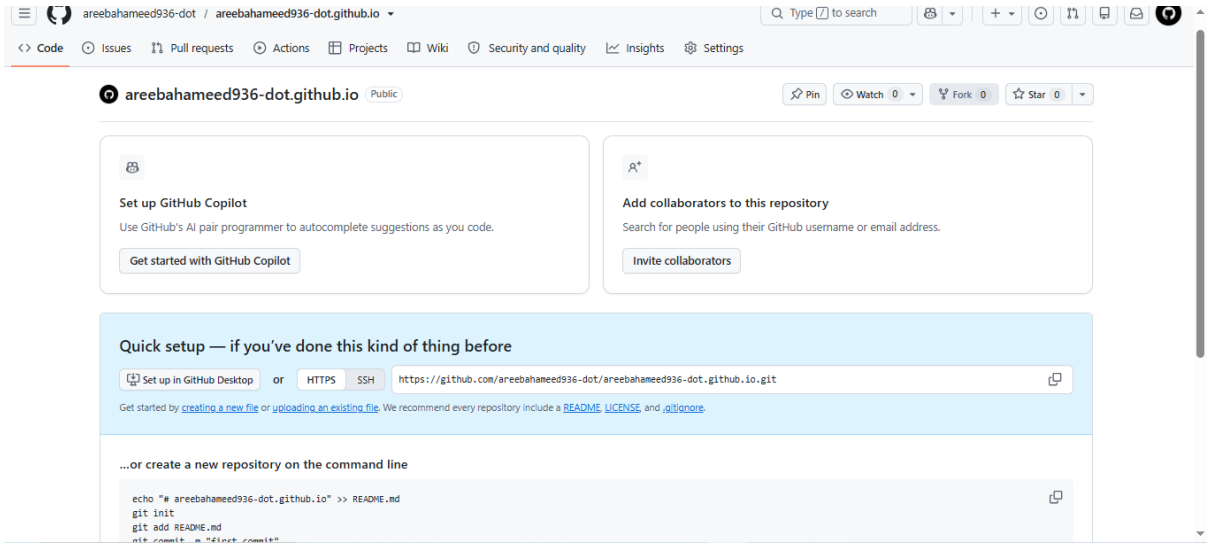
gitea_dashboard.png



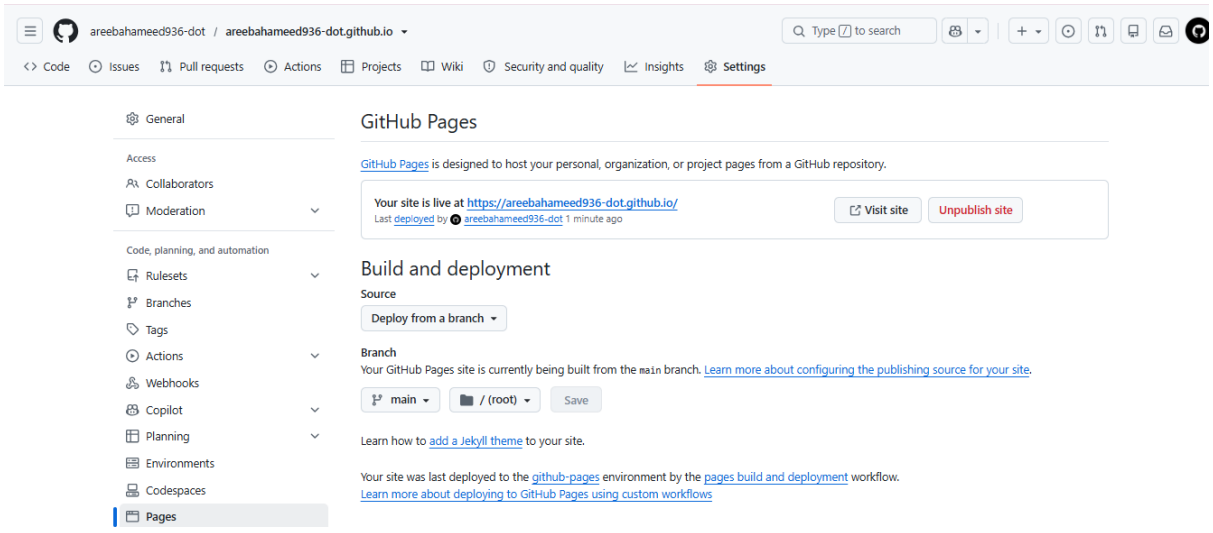
gitea_new_repo.png



github_pages_repo.png



github_pages_settings.png



Live_site.png

